Caleb A. Shaw

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Education

North Carolina State University, Raleigh, NC 2022–Expected May 2027
Ph.D., Nuclear Engineering, minor in Mathematics

- Dissertation (working title): "Hybrid Stochastic–Deterministic Methods Based on Coupled Moment Equations for Linear Transport Problems." Advisor: Dmitriy Anistratov.
- Master of Nuclear Engineering (2025), GPA 3.72.

University of New Mexico, Albuquerque, NM 2022
B.S., Nuclear Engineering, minor in Mathematics. GPA 3.73, Cum Laude

Research Experience

North Carolina State University, Raleigh, NC Aug. 2022–Present
Graduate Research Assistant, CARRE and CEMeNT (DOE/NNSA PSAAP centers); advisor Dmitriy Anistratov

- Designed and implemented an automatic weight window method for global variance reduction in time-dependent Monte Carlo transport. Window centers come from an auxiliary hybrid MC/deterministic low-order second-moment (LOSM) solve, discretized to second order in time and space, which keeps the particle population spatially uniform. Quantified efficiency with figure of merit and relative error against an analytic benchmark [J1, C2].
- Developing two-level hybrid macro–micro transport methods that split the solution into a deterministic macro component and a Monte Carlo micro component. Extended the framework with quasi-Monte Carlo sampling and maximum-entropy closures [C1, S1, S2].
- Contributing algorithm and code development to the new-science branch of CARRE's software.
- Developer on MC/DC, an open-source, HPC-oriented Python Monte Carlo transport code built with national laboratory and university collaborators; implemented its weight window methods [J2].
- Designed and prototyped MC/DC's domain decomposition algorithm, removing the memory bottleneck that limited problem size in massively parallel runs and enabling simulations on LLNL supercomputers (Dane, Quartz, Tuolumne) at up to 512 nodes.
- Applied dynamic mode decomposition to Monte Carlo tallies to predict flux for weight window generation.
- Presented results at Tri-Lab Support Team meetings, DOE/NNSA annual reviews, and center symposia (2023–2026); contributed to CARRE's 2026 annual progress report.

Los Alamos National Laboratory, Los Alamos, NM Summer 2023
Graduate Student Intern, XTD Division

- Generated and processed radiation hydrodynamics data to train compressed sensing models (with Dr. Shane Coffing).
- Modified input decks and ran full 3D simulations extending existing 2D r–z problems (mentor: Dr. Todd Urbatsch).

Los Alamos National Laboratory, Los Alamos, NM Summer 2022
Graduate Student Intern, T-3 Division

- Implemented the method of manufactured solutions in a radiation hydrodynamics code (mentor: Dr. HyeongKae Park).
- Performed and presented spatial and temporal error analyses to verify the code.

University of New Mexico, Albuquerque, NM Fall 2021–Spring 2022
Undergraduate Research Assistant, Optimization and Controls Lab (advisor: Claus Danielson)

- Researched optimization techniques for engineering applications in ongoing lab projects.

Publications

Journal articles

[J1] **C. A. Shaw** and D. Y. Anistratov, "Hybrid Weight Window Method for Global Time-Dependent Monte Carlo Particle Transport Calculations," *J. Comput. Theor. Transp.*, 2026. doi:10.1080/23324309.2026.2638248

[J2] J. P. Morgan et al. (incl. **C. A. Shaw**), "Monte Carlo / Dynamic Code (MC/DC): An Accelerated Python Package for Fully Transient Neutron Transport and Rapid Methods Development," *J. Open Source Softw.*, 9(96):6415, 2024.

Conference papers and transactions

[C1] **C. A. Shaw** and D. Y. Anistratov, "Hybrid Two-Level Transport Method with Solution Decomposition in Macro and Micro Components," *Trans. Am. Nucl. Soc.*, 135, 2026 (to appear). arXiv:2607.01346

[C2] **C. A. Shaw** and D. Y. Anistratov, "Hybrid Weight Window Techniques for Time-Dependent Monte Carlo Neutronics," *Proc. M&C 2025*, Denver, CO, 2025. arXiv:2501.06144

Submitted

[S1] **C. A. Shaw** and D. Y. Anistratov, "A Two-Level Hybrid Quasi-Monte Carlo / Deterministic Method for Particle Transport with Variance Reduction," submitted to M&C 2027.

[S2] **C. A. Shaw** and D. Y. Anistratov, "Hybrid Macro-Micro Method with Maximum Entropy Approximation for Particle Transport Problems," submitted to M&C 2027.

Presentations

- "Hybrid Two-Level Transport Method with Solution Decomposition in Macro and Micro Components," oral presentation, ANS Winter Conference, Phoenix, AZ, Nov. 2026 (upcoming).
- "Hybrid Micro-Macro Methods for Particle Transport," oral presentation, CARRE Research Symposium, 2026.
- "A Two-Level Hybrid Quasi-Monte Carlo / Deterministic Method for Particle Transport," poster, CARRE Annual Review, 2026.
- "Hybrid Weight Window Techniques for Time-Dependent Monte Carlo Neutronics," oral presentation, M&C 2025, Denver, CO, Apr. 2025.
- "Hybrid Weight Windows in MC/DC," CEMeNT Research Meeting, Aug. 2024.
- "Domain Decomposition in MC/DC," Tri-Lab Support Team Meeting, Portland, OR, Apr. 2024.
- "Analysis of Dynamic Mode Decomposition and Flux Prediction for Weight Windows," poster, Tri-Lab Support Team Meeting, Portland, OR, Apr. 2024.

Professional Service

- Peer reviewer, *Nuclear Science and Techniques*, 2026–present.

Technical Skills

- **Languages:** Python (NumPy, SciPy, pandas, Matplotlib, Numba, mpi4py), C++ (MPI, OpenMP)
- **HPC:** MPI-parallel codes, domain decomposition, runs up to 512 nodes on LLNL systems (Dane, Quartz, Tuolumne); SLURM and other schedulers; workflows on LANL and NCSU systems
- **Methods:** Monte Carlo and quasi-Monte Carlo, variance reduction (weight windows), hybrid MC/deterministic and moment methods, implicit Monte Carlo, radiation hydrodynamics, nonlinear PDEs, stochastic processes, optimization
- **Codes:** MCNP, SCALE, MC/DC, CASSIO, Osito
- **Tools:** Git/GitHub, ParaView, LaTeX

Relevant Coursework

- **Math and numerics:** Numerical Nonlinear PDEs, Numerical Analysis, Nonlinear Optimization, Probability and Stochastic Processes
- **Physics and nuclear:** High Energy Density Physics, Reactor Design, Plasma Physics, Computational Transport, Atomistic Simulations, Nuclear Fuel Cycle

Honors and Leadership

- Treasurer, Pi Kappa Alpha, University of New Mexico, 2021–2022: managed the chapter budget and led fundraising that raised several thousand dollars.
- 1st Place, Pitch Contest, UNM Engineering Expo VII (2022); Academic Achievement Award, UNM (2021); Alpha Nu Sigma Honor Society (2022)